Gourishetty Himani

ai24btech11011@iith.ac.in | himanigourishetty12@gmail.com | +91 8247765460 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.73
sri chaitanya junior college	Class XII (Telangana state board of intermediate education)	2024	98.50%
Paramita heritage school	Class X (Central board of secondary examination)	2022	96.60%

SKILLS

programming languages: c , c++ , python

Tools & Technologies : Latex, MySQL

Soft Skills: Teamwork, Communication, Leadership

RELEVANT COURSES

Artificial Intelligence: Introduction to AI, Programming for AI, Foundations of Machine Learning, Deep Learning, AI For Humanity

Computer Science: Introduction to Programming, Data Structures & Applications, DBMS-1, Algorithms

Mathematics: Probability & Random Variables, Vector Calculus, Discrete Mathematics, Calculus-I, Calculus-II, Matrix Theory , Differential Equations, Series of Functions, Numerical Analysis, Introduction to Metric Spaces

Other: linear systems and signal processing, Control Systems

PROJECTS

FailSafe - Student Failure Prediction System | (Link)

- Developed an AI-powered web application to predict students at risk of academic failure using an **XGBoost** classification model.
- Integrated **SHAP** explainability to identify key risk factors and provide interpretable predictions with personalized intervention recommendations.
- Built role-based faculty and student dashboards with **CSV batch upload**, prediction visualization, and risk monitoring using **Django**

Fair Representation Learning| (Link)

- Collaborated on a project exploring fairness-aware representation learning using **Variational Autoencoders, Information Bottleneck**, and **Graph Neural Networks.**
- Implemented and evaluated the **Flexible Fair Variational Autoencoder (FFVAE)** to learn fair latent representations while balancing predictive performance and fairness.
- Conducted experiments to analyze the trade-off between representation fairness and downstream predictive utility.

Custom CNN for Image Classification

- Designed and trained a custom convolutional neural network with **Residual Blocks**, **Squeeze-and-Excitation (SE) attention**, and **Batch Normalization** for image classification.

- Applied data augmentation, AdamW optimization, and hyperparameter tuning to improve validation performance.
- Built an end-to-end inference pipeline to generate predictions on unseen test images and export submission-ready results.
- Achieved **20th place out of 74 teams** in the course competition.

Classical Machine Learning Algorithms

- Built PCA, K-Means, Gaussian Mixture Models (EM), Polynomial Regression, and MLE/MAP models from scratch.
- Applied these algorithms to face recognition, image generation, clustering, and regression tasks.
- Evaluated model performance using reconstruction error, clustering metrics, and predictive accuracy

EXTRACURRICULAR

Actively participated in walkathons, swatch bhaarat and vidyadaan events organised bt NSS team

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Probability Course| (Jan 2026 - May 2026)

Assisted students with coursework and doubt resolution in Probability; supported tutorials and academic activities.

Diesta Event Coordinator |(Jan 2026 - May 2026)

Coordinated event logistics and participant management; collaborated with teams to ensure smooth event execution